Triangle Similarity — Reversed Correspondence and Scale-Factor Errors

Answer Key

High School Geometry

Quick Answers

1. 12	3. 6	5. 27.5
2. 21	4. 128	6. 12

Curriculum

Level: High School Geometry
HSG-SRT.A.1–A.3, HSG-SRT.B.4–B.5 – *problems 1, 2, 3, 4, 5, 6*
Difficulty 1–5 of 5 across 7 tagged checks.

Common wrong answers

2. *If they got 9.33: wrote the scale factor upside down, shrinking a side of the larger triangle*
3. *If they got 13.5: multiplied by the large-over-small factor while moving to the smaller triangle*
4. *If they got 80: scaled the area by the side ratio instead of by its square*
5. *If they got 17.6: paired the flagpole's shadow with the person's height, reversing the correspondence*
6. *If they got 33.33: used the large-over-small scale factor to find a side of the smaller triangle*

1. $\triangle ABC \sim \triangle DEF$, $AB = 6$, $BC = 8$, $DE = 9$. Find EF.

Step 1: read the correspondence off the statement: $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$, so $\overline{AB}$ pairs with $\overline{DE}$ and $\overline{BC}$ pairs with $\overline{EF}$. Step 2: decide the direction. $AB = 6$ grows to $DE = 9$, so DEF is the larger triangle and EF must come out larger than $BC = 8$. Step 3: set up one proportion, both ratios running small-to-large: $\dfrac{6}{9} = \dfrac{8}{EF}$. Step 4: cross-multiply: $6 \cdot EF = 72$, so $\boxed{EF = 12}$. Step 5: check against step 2 — $12 > 8$ ✓, and $\dfrac{6}{9} = \dfrac{8}{12} = \dfrac{2}{3}$ ✓.

2. Find and fix: Kai's TU for $\triangle PQR \sim \triangle STU$, $PQ = 10$, $QR = 14$, $ST = 15$.

Step 1: check each of Kai's ratios for direction. $\dfrac{ST}{PQ}$ runs small-to-large is $\dfrac{15}{10}$: large over small. His other ratio $\dfrac{QR}{TU}$ runs from triangle PQR to triangle STU: small over large. Step 2: name the error — the two ratios point in opposite directions, so his equation says the large triangle is both bigger and smaller than the small one. Written consistently it must be $\dfrac{ST}{PQ} = \dfrac{TU}{QR}$. Step 3: since $ST = 15$ is longer than $PQ = 10$, triangle STU is the larger one and TU must exceed $QR = 14$ — Kai's 9.33 is already impossible. Step 4: solve the corrected proportion: $TU = 14 \cdot \dfrac{15}{10} = \boxed{21}$. Step 5: check: $\dfrac{15}{10} = \dfrac{21}{14} = 1.5$ ✓.

3. Find and fix: $\triangle ABC \sim \triangle FED$, $AB = 12$, $BC = 9$, $FE = 8$. Find ED.
Step 1: read the correspondence in the order written: $A \leftrightarrow F$, $B \leftrightarrow E$, $C \leftrightarrow D$. So $\overline{AB}$ pairs with $\overline{FE}$ and $\overline{BC}$ pairs with $\overline{ED}$ — Rho's pairing is right. Step 2: find the direction. $AB = 12$ corresponds to $FE = 8$, so the second triangle is *smaller*: the scale factor from ABC to FED is $\dfrac{8}{12} = \dfrac{2}{3}$, a number less than 1. Step 3: name the error — Rho used $\dfrac{12}{8}$, the factor for going the other way. Multiplying by a factor greater than 1 makes the side of the smaller triangle longer than the side it came from, which cannot happen. Step 4: $ED = 9 \cdot \dfrac{8}{12} = \boxed{6}$. Step 5: check: $\dfrac{8}{12} = \dfrac{6}{9}$ ✓, and $6 < 9$ ✓.

4. Sides in ratio $5 : 8$, smaller area 50 square centimetres. Find the larger area.
Step 1: name what Rae scaled incorrectly — she applied the *side* factor $\frac{8}{5}$ to an *area*. Lengths scale by k; areas scale by k^2, because an area is a product of two lengths and each one is multiplied by k. Step 2: the linear factor from smaller to larger is $k = \dfrac{8}{5} = 1.6$, so the area factor is $k^2 = \dfrac{64}{25} = 2.56$. Step 3: multiply: $50 \times 2.56 = \boxed{128}$ square centimetres. Step 4: check the size: 128 is more than twice 50, which is right for sides that grew by 60%; Rae's 80 is only 1.6 times the original area.

5. Find and fix: person 5 ft with a 4 ft shadow, flagpole with a 22 ft shadow. Find the flagpole's height h.
Step 1: state the correspondence in words. In each right triangle, the height is one leg and the shadow is the other, so heights correspond to heights and shadows correspond to shadows. Step 2: locate Ines's error. In her $\dfrac{5}{4} = \dfrac{22}{h}$, the left side is height over shadow, but on the right she put the flagpole's *shadow* on top and its height underneath — shadow over height. She paired the flagpole's shadow (22) with the person's height (5). Step 3: write it with both ratios in the same order, height over shadow: $\dfrac{5}{4} = \dfrac{h}{22}$. Step 4: solve: $h = 22 \cdot \dfrac{5}{4} = \boxed{27.5}$ ft. Step 5: check the scale: the flagpole's shadow is 5.5 times the person's, so its height must be 5.5 times 5 ft ✓. Ines's 17.6 ft is barely three times the person's height while the shadow is over five times as long.

6. Challenge: $\triangle JKL \sim \triangle MNP$, $JK = 15$, $MN = 9$, $KL = 20$. Diagnose Tomas's NP and find the correct value.

Parts (a) and (b) are open explanations — flagged for manual review. Model answers below; part (c) is machine-verified.

(a) $JK = 15$ corresponds to $MN = 9$, so $\triangle MNP$ is the *smaller* triangle. Every side of MNP must therefore be shorter than the side of JKL it matches. NP corresponds to $KL = 20$, so NP has to be less than 20 — Tomas's 33.33 is longer, so it is impossible no matter what the arithmetic says.

(b) A usable rule: put the side of the triangle you are going *to* on top, and its matching side in the triangle you are coming *from* underneath:

$$k = \frac{\text{going to}}{\text{coming from}}$$

Going to the smaller triangle that fraction is less than 1; going to the larger it is greater than 1. Decide which before multiplying, then check the answer against that decision.

(c) Step 1: the factor from JKL to MNP is $\dfrac{MN}{JK} = \dfrac{9}{15} = 0.6$, which is less than 1 as part (a) requires. Step 2: $NP = KL \cdot \dfrac{9}{15} = 20 \cdot 0.6 = \boxed{12}$. Step 3: check: $\dfrac{9}{15} = \dfrac{12}{20}$ ✓, and $12 < 20$ ✓. Tomas's 33.33 is exactly $20 \div 0.6$ — the signature of a scale factor used upside down.

Full credit on (a) and (b) needs: MNP named as the smaller triangle, with the pair 15 and 9 given as the reason; the conclusion that NP must be under 20; and a rule that fixes the *direction* of the factor, not a restatement of the arithmetic.